Manish Mahendra More

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.81
Shri H. J. Gajera Higher Secondary English School	Class XII (GSHSEB)	2020	70.70%
Little Flower School, Loni	Class X (CBSE)	2018	84.80%

SCHOLASTIC ACHIEVEMENTS

- **Academic Excellence Award** : Secured Highest CGPA in MTech Smart Mobility (2025 batch)
- **Research Contribution Award** : Awarded for leading UAV autonomy and navigation research at **TiHAN.**

- **NvisionX Hackathon** : Secured 3rd place for an **IoT-based fleet optimization solution** for Hyderabad City.

RESEARCH EXPERIENCE

TiHAN - Technology Innovation Hub on Autonomous Navigation, IIT Hyderabad

India's first autonomous navigation testbed for terrestrial and aerial systems.

TiHAN x Mitsubishi Electric | Developing a UAV-based structural inspection system for automated crack detection.

- Building **segmentation and classification pipelines** for crack localization and severity estimation.
- Estimating crack dimensions via monocular depth estimation, camera calibration, and image-based measurement.
- Integrating **ROS2, OpenCV, and AI perception** for autonomous infrastructure monitoring.

TiHAN × MELCO | Adaptive Distributed Autonomy Framework for Indoor UAV Navigation.

- Developing a **GNSS-denied UAV** with **LiDAR, vision,** and **Jetson** edge computing.
- Designing **semantic mapping** and **adaptive task allocation** across UAV and orchestrator.
- Building an intelligent **inspection pipeline** for anomaly detection and resilient operation.

PROJECTS

Curse of Dimensionality in MCMC Sampling

Empirical study of dimensionality effects on the efficiency and accuracy of MCMC sampling algorithms.

- Benchmarked five MCMC algorithms on Gaussian posteriors from 1-50 dimensions using **PyMC and NumPyro.**
- Evaluated ESS/s, KL divergence, posterior error, R, and runtime across dimensions.
- Found NUTS most accurate while emcee achieved the highest throughput on simpler targets.

OpenAirInterface 5G Network Deployment & Performance Evaluation

End-to-end LTE and 5G SA deployment and benchmarking for next-generation wireless networks.

- Deployed LTE and 5G SA networks with **OAI, Docker, Kubernetes, WSL2,** and **Minikube**.
- Configured 5GC, gNB, UE, CU/DU split, BubbleRAN, and RF simulator for network experiments.
- Benchmarked **LTE vs 5G NR** on throughput, latency, QoS, numerology, and multi-UE performance.

Fleet Management for Waste Collection (NvisionX Hackathon)

Intelligent waste-collection platform combining dynamic fleet routing with IoT-based demand estimation.

- Built dynamic fleet routing using real-time traffic and demand-based scheduling.
- Integrated IoT bin-fill sensors to prioritize collection based on predicted demand.

- Reduced unnecessary trips and estimated fuel use through demand-driven dispatch.

LocalMind: Autonomous AI Research Agent

Local-first multi-agent system for research automation, knowledge retrieval, and experiment management.

- **Built** a local AI research agent using Qwen/DeepSeek for paper analysis and workflow automation.
- **Integrated** RAG, SQL, and vector memory for contextual retrieval and long-term knowledge retention.
- **Automated** experiment tracking, statistical analysis, and report generation using agentic workflows.

Elephant & Rabbit Robots - ABU Robocon

Competition robotics project involving the design, development, and integration of two competition robots.

- Led design and development of two competition robots for ABU Robocon 2023.
- Coordinated a 23-member team and integrated Arduino, motor drivers, pneumatics, and mechanisms.
- Achieved AIR-14 with a combined score of 96/100 in national Robocon qualification rounds.

Portfolio Website : manizz138.github.io

TECHNICAL SKILLS

- **Programming :** C, C++, Python, MATLAB
- **AI & Robotics :** ROS2, Nav2, MoveIt2, Gazebo, RViz, OpenCV, TensorFlow, PyTorch, YOLO , Huggingface
- **Networking & Cloud :** Docker, Kubernetes, OpenAirInterface (OAI), BubbleRAN, NS-3
- **Embedded Systems :** ESP32, Arduino, KiCAD, IoT
- **AI Tools :** ChatGPT, Claude, Claude Code, Gemini, Antigravity, Cursor, Emergent, Ollama
- **Other :** Linux, Git, LATEX

POSITIONS OF RESPONSIBILITY

- **Team Captain** ABU Robocon 2024
- **PR and Finance Core** - Google developer students clubs.
- **Senior Mechatronics Core** - Tinkerers lab IIT Hyderabd
- **Hospitality coordinator** - Inter IIT sports meet 2025

HOBBIES AND INTRESTS

- **Hockey:** Goalkeeper for Inter-IIT Hockey Team.
- **Gaming:** Competitive FPS games including Valorant and Counter-Strike 2; PlayStation gaming.
- **Content Creation:** Create informational content for students through personal social media and YouTube.
- **Swimming:** Inter-IIT Probables; Gold Medalist - 50m Freestyle Relay; Silver Medalist - 50m Freestyle.